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Training Humanoid Robots to Walk in Lunar Gravity Using Reinforcement learning

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Abstract

Humanoid robots are promising candidates for lunar exploration, but adapting locomotion to reduced gravity remains a challenge. This paper investigates the use of reinforcement learning to train a Unitree G1 humanoid to walk in simulated lunar gravity using NVIDIA Isaac Sim 5.0 and Unitree RL Lab. Three policies were developed: an Earth baseline, a fine-tuned lunar model adapted from the Earth policy, and a lunar-from-scratch model. Results show that lunar policies autonomously discovered gait adaptations, including shortened strides, toe-dominant walking, and a forward-shifted centre of mass, consistent with prior low-gravity analyses. Fine-tuning proved more efficient, achieving performance comparable to training from scratch in less than half the time. These findings demonstrate that reinforcement learning can effectively adapt humanoid locomotion to extraterrestrial conditions and highlight fine-tuning as a practical strategy for planetary robotics missions.

1 Introduction

Humanoid robots will be vital in the future of lunar and space exploration, because of their adaptability and human-inspired form. The work reported here aims to leverage the advanced simulation software with reinforcement learning to investigate the difficulties of developing and refining advanced locomotion models of humanoid robots in lunar gravity.

The potential of humanoid robots in space has been recognised for over two decades by the space industry, primarily due to their anthropomorphic design (Stoica et al., 2005). This recognition is demonstrated by significant, high-profile investments from both public and private sectors. NASA, for instance, is actively developing its Valkyrie robot to operate in human-engineered environments and perform dangerous extravehicular tasks,



Figure 1: A humanoid robot in a Moon habitat module.

aiming to support future extraterrestrial spaceflight endeavours (Radford et al., 2015). Similarly, SpaceX has committed to its Optimus program, unveiled at its 2022 AI Day, with the stated goal of eventually sending its robots to assist in building a colony on Mars (Tesla, 2022).

The core advantage of this human-like form is twofold. First, their dexterous, human-like hands allow them to operate tools and equipment already designed for astronauts. This capability is critical as it drastically reduces the mission payload and logistical burden by eliminating the need to send heavy, specialised robotic equipment. Second, their bipedal structure enables them to navigate diverse terrain, ranging from the human-sized catwalks,

scaffolding, and ladders found in space habitats (Stolica et al., 2005) to complex and unstructured planetary surfaces. These combined capabilities make them ideal for preparatory tasks—such as assembling habitats or accessing confined spaces prior to human arrival (Omer et al., 2011)

In this work, we use NVIDIA Isaac Sim 5.0 and the Unitree RL Lab to investigate how reinforcement learning can be applied to adapt humanoid locomotion to lunar gravity. A Unitree G1 humanoid robot model was trained under three conditions: (i) a baseline policy trained in Earth gravity, (ii) a fine-tuned lunar policy adapted from the Earth baseline, and (iii) a lunar-from-scratch policy. These experiments were designed to evaluate how gravity affects walking gait, centre of mass positioning, and control accuracy, while also comparing the efficiency of transfer learning against training from scratch. The results provide new insights into how modern reinforcement learning methods can support humanoid deployment in reduced-gravity environments.

This paper makes the following contributions:

- The first demonstration of reinforcement learning-based humanoid locomotion in simulated lunar gravity using the Unitree G1 model.
- A comparative study of Earth-baseline, lunar fine-tuned, and lunar-from-scratch policies, revealing distinct gait adaptations.
- Empirical evidence that fine-tuning reduces training time by more than half while achieving equivalent performance.
- Validation of prior analytical insights on low-gravity walking behaviours using a modern high-fidelity simulator.
- Extension of reinforcement learning research on humanoid locomotion to extraterrestrial environments, providing a foundation for future planetary exploration.

2 Background

2.1 The Choreography of Walking

Bi-pedal walking is the form of locomotion in which the body alternates between single-leg support and double support, requiring a constant interplay of balance, forward momentum, and ground reaction forces. The primary role of gravity in walking is to allow pendular energy exchange, where gravitational potential energy converts to kinetic energy as the centre of mass (COM) descends, reducing the external work required of the muscles to maintain motion, while also allowing the generation of frictional forces and stabilising each step (Cavagna et al., 2000). Hence, this also significantly affects humanoid robot locomotion, as demonstrated by

studies analysing walking gait and velocity under varying gravitational conditions (Omer et al., 2011, 2014). In environments of reduced gravity such as the Moon (1/6th Earth’s gravity) or Mars (1/3rd Earth’s gravity), lower gravitational forces demand slower walking speeds to maintain balance or, alternatively, switch to running gaits to achieve the necessary dynamic stability (Omer et al., 2011).

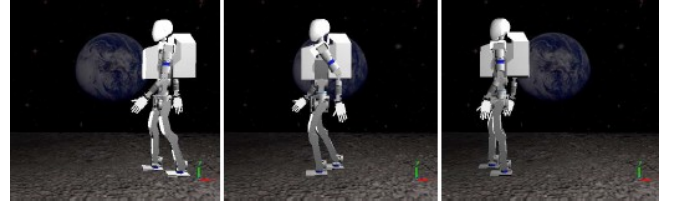


Figure 2: Early Simulation of a Humanoid Robot on the Moon (Omer et al., 2014).

For example, simulations in Omer et al. (2014) show that a humanoid robot on the Moon would need to reduce its walking speed by approximately 60% compared to Earth to avoid tipping or adopt a hopping motion similar to a run (Omer et al., 2011). Furthermore, changes in hip height and acceleration during a step cycle illustrate how reduced gravity diminishes downward force, affecting friction generation and forward propulsion (Omer et al., 2014).

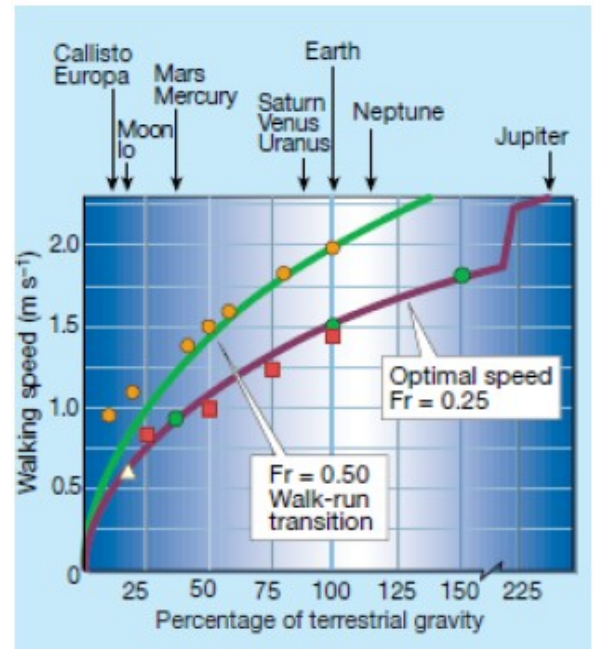


Figure 3: Max Stable Velocity Vs Gravity (Omer et al., 2011).

The total effect of gravity on the maximum stable

walking speed is seen in Figure 3, highlighting the need to adapt modern advanced control systems and gait strategies tailored to specific gravitational contexts, ensuring effective locomotion on extraterrestrial surfaces.

2.2 Training Humanoids via Simulation

Given the large numbers of degrees-of-freedom of humanoid robots and the subsequent difficulty in developing both robust joint control and walking gaits using real robots, there has been a tradition of using simulation to solve the humanoid walking issue. Kee et al. (2004) proposed a two-stage, biologically inspired control scheme for the GuRoo humanoid robot. First, a Genetic Algorithm tuned local PI feedback controllers to minimise error and vibration; then a cerebellar-inspired neural network (CMAC) provided feed-forward compensation that learned online from trajectory error. Experiments, including crouching motions, showed the benefit of combining evolutionary tuning with adaptive neural control.

The MuJoCo physics engine (Multi-Joint dynamics with Contact) was introduced by Todorov et al. Todorov et al. (2012) to provide a fast and accurate platform for simulating continuous, contact-rich dynamics in robotic and biomechanical systems. Unlike graphics-oriented engines, MuJoCo was designed from the outset to support model-based control and reinforcement learning research, with a focus on efficiency, differentiability, and stable contact handling.

Since its release in 2012, MuJoCo has become a standard simulation environment for developing and benchmarking reinforcement learning algorithms. It underpins widely used environments such as OpenAI Gym’s Humanoid, Ant, and HalfCheetah tasks, as well as the DeepMind Control Suite Tassa et al. (2018). These benchmarks have been instrumental in driving advances in continuous control. In 2021, MuJoCo was acquired by DeepMind and released as open source DeepMind (2021), significantly broadening its accessibility to the robotics and AI research communities.

2.3 Leveraging Cutting Edge Technologies

Nvidia Isaac Sim 5.0

NVIDIA Isaac Sim 5.0 is a high-fidelity robotics simulation platform built to emphasise accurate physics, extensibility, and open-source adaptability. It uses the PhysX engine under the hood and supports Universal Scene Description (USD) schemas for physics objects and environment assets, enabling simulations with realistic rigid body dynamics, multi-joint articulation, sensor (camera, LiDAR, contact) simulation, and advanced features like friction models, velocity/torque limits, and deformables (in beta) (NVIDIA Corporation, 2025d). Because much of the Isaac Sim codebase, including Isaac Sim-specific

extensions, is now open source under the Apache 2.0 license, users can adapt, extend or integrate it into custom robotics or simulation workflows (Isaac Sim Development Team, 2025)

System Requirements

Element	Minimum Spec	Good	Ideal
OS	Ubuntu 22.04/24.04 Windows 10/11	Ubuntu 22.04/24.04 Windows 10/11	Ubuntu 22.04/24.04 Windows 10/11
CPU	Intel Core i7 (7th Generation) AMD Ryzen 5	Intel Core i7 (9th Generation) AMD Ryzen 7	Intel Core i9, X-series or higher AMD Ryzen 9, Threadripper or higher
Cores	4	8	16
RAM	32GB [1]	64GB [1]	64GB [1]
Storage	50GB SSD	500GB SSD	1TB NVMe SSD
GPU	GeForce RTX 4080	GeForce RTX 5080	RTX PRO 6000 Blackwell
VRAM	16GB [1]	16GB [1]	48GB

Figure 4: NVIDIA Isaac Sim 5.0 System Requirements (NVIDIA Corporation, 2025b).

However, there are some trade-offs. Firstly, the hardware performance requirements are steep, requiring users to have compute specified by NVIDIA (Figure 4) and forcing users to own only NVIDIA graphics cards (NVIDIA Corporation, 2025b). Secondly, as a relatively new release, the installation is complex, documentation and ecosystem support are marginal, some parts less mature or have fewer examples/tutorials, and known issues are frequently being discovered (NVIDIA Corporation, 2025c). This can make working with the simulator difficult and requiring solving novel problems that have yet to be discussed online. Overall, Isaac Sim 5.0 is a strong step forward for physically accurate, extensible robotics simulation, especially for those able to meet the hardware demands and willing to work with evolving documentation and APIs. This simulator was chosen for the project because it’s the most realistic and most advanced opensource software that is available, along with its seamless integration with Isaac Lab, a reinforcement learning extension for Isaac Sim, which allows users to use reinforcement learning to train robots new task.

Nvidia Isaac Lab

NVIDIA Isaac Lab is an open-source, unified and modular framework for robot learning, designed to streamline common workflows in robotics research such as reinforcement learning, imitation learning, and motion planning, built on top of NVIDIA Isaac Sim. It aims to reduce the sim-to-real gap and accelerate development of robot policies by leveraging GPU-accelerated physics simulation through PhysX and RTX based rendering, enabling massively parallel execution of environments, which combines accurate sensor simulation, domain randomisation, and support for multiple robot embodiments

Table 1: Comparison of recent RL approaches for humanoid locomotion in MuJoCo and related simulators.

Paper	Robot / Env	Method	Key Contribution	Relevance to Lunar Gait Study
Singh et al. (2022) <i>Learning Bipedal Walking on Foot-planned steps</i>	MuJoCo humanoid (HRP5P, JVRC-1)	RL with curriculum learning + footstep planner	Robust locomotion including turning, stairs, uneven terrain using footstep conditioning	Shows how curriculum + task shaping improves generalisation, relevant for uneven lunar surfaces
Seo et al. (2025) <i>FastTD3</i>	MuJoCo Humanoid benchmark	FastTD3 (improved TD3 variant)	Accelerates RL training with large batches, parallel sim, distributional critic	Demonstrates how efficiency can be doubled—aligns with fine-tuning vs scratch training findings
Sferrazza et al. (2024) <i>HumanoidBench</i>	MuJoCo Unitree H1 + hands	RL benchmarks (PPO, SAC, etc.)	Broad benchmark for locomotion + manipulation in simulation	Provides community baseline; this work adds low-gravity locomotion as a new challenge
This Work (2025) <i>Unitree G1 in Lunar Gravity</i>	Isaac Sim + Unitree G1	RL (PPO via Unitree Lab)	First study of humanoid RL locomotion in reduced gravity; fine-tuning halves training time	Extends RL humanoid work into extraterrestrial environments—novel domain shift

(manipulators, quadrupeds, humanoids, etc.) A large range of ready-to-use example environments, tasks, sensors and tools that use detailed robot models from leading robotics companies such as Unitree, Boston Dynamics, Franka Emika, and Universal Robots are provided, along with the ability to customise or build new ones, emphasising modularity and extensibility. The framework supports running locally or in the cloud, enabling scalable experiments across many robot configurations (NVIDIA Corporation, 2025a). Isaac Lab’s ability to leverage parallel computing across multiple environments, its support for industry robots such as the Unitree G1, and the modularity of its environments, combined with a plethora of ready-to-go examples, made it an ideal choice for the project. These features enabled faster learning of the software and reduced overall code complexity by abstracting the modelling, simulation, and reinforcement learning components.

Unitree Robotics

The Unitree G1 (Figure 5) is a full-sized humanoid robot developed by Unitree Robotics. It stands approximately 127 cm tall, weighs 35 kg, and features 23 degrees of freedom, including actuated arms, legs, and a torso. The G1 is designed for general-purpose research in locomotion, manipulation, and human–robot interaction. It is powered by high-torque actuators, has integrated sensing for balance and motion control, and can operate untethered with onboard batteries. Its anthropomorphic design, compact form factor, and availability to research

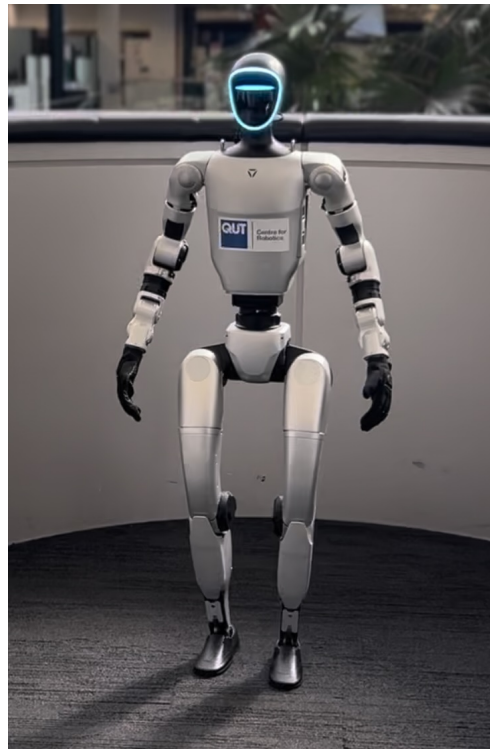


Figure 5: Unitree G1 humanoid robot.

institutions make it a practical platform for developing and evaluating humanoid control policies.

Unitree state that they are committed to foster-

ing innovation through open-source resources, notably the Unitree RL Lab, a comprehensive repository built on IsaacLab that provides reinforcement learning environments for robots like the G1. This open-source project enables researchers and developers to train, simulate, and deploy policies, accelerating advancements in robotic autonomy and intelligence (Unitree Robotics, 2024). The G1’s availability and open source software tools make this ecosystem an ideal pick for our work.

Foundation Humanoid Robot Models

Foundational humanoid robot models such as NVIDIA Isaac GR00T are large, pre-trained artificial intelligence systems developed to provide generalised reasoning, perception and control capabilities across a range of humanoid robot embodiments and tasks. They are used to accelerate robotics research and deployment by enabling robots to understand high-level instructions (often including language and vision), plan, adapt to new environments, and execute continuous control policies without needing from-scratch training for every new task or body (NVIDIA Corporation, 2025a,b). Examples of such models include Isaac GR00T N1 / N1.5 from NVIDIA, which supports multimodal inputs and is fine-tuned for different embodiments and tasks. These models are trained on large and diverse datasets comprising real robot demonstrations, synthetic trajectories (generated in simulation), internet-scale video, plus multimodal data pairing observations, actions, and often language or high-level descriptions (Liu et al., 2024). Training regimes often employ pretraining on broad, generic tasks/environments followed by fine-tuning for specific embodiments or tasks (NVIDIA Corporation, 2025b). The biggest challenges faced by foundational models include scarcity of robot-relevant data especially with realistic physics and safety annotations, the difficulty of generalisation across different robot embodiments and environments, ensuring safe and reliable behaviour in real-world deployments, dealing with uncertainty, and achieving real-time performance (Liu et al., 2024; NVIDIA Corporation, 2025a). Overall, foundational humanoid models represent a critical step toward developing generalist robotic systems, but their success will depend on overcoming data, safety, and deployment challenges.

Reinforcement Learning Policy

Reinforcement Learning (RL) trains robots through a trial-and-error process, where they learn optimal actions by receiving rewards or penalties based on task performance. This approach excels in dynamic environments, enabling robots to refine behaviours without explicit programming (Chen et al., 2020). For example, RL has been applied to enhance walking stability in humanoid robots by optimizing gait patterns. (Chen et al., 2020) describe a study where RL was used to train a robot

to adjust its step length and timing, improving its ability to traverse uneven terrain and recover from external perturbations like pushes or slips. Despite their effectiveness, RL policies face challenges including high computational cost, difficulty generalising beyond trained tasks, and ensuring safe, robust performance in unpredictable real-world conditions (Kober et al., 2013). Both RL and foundational models enhance humanoid robot capabilities, but they face distinct challenges. RL is limited to a task specific scope, while foundational models are limited by the quantity and diversity of human demonstrations (Chen et al., 2020; Gu et al., 2025). Current research aims to address these issues by combining the two through synthetic datasets generated in physics simulators are being used to augment training, reducing dependency on real-world data (Gu et al., 2025).

2.4 Knowledge Gap

Previous studies have explored humanoid locomotion in lunar gravity through simplified simulation models (Omer et al., 2011; Stoica et al., 2005), but these approaches have not been reproduced using modern humanoid platforms or contemporary reinforcement learning methods. Early work relied heavily on analytical control systems and mathematical models to adapt motor commands across different gravities (Omer et al., 2014), whereas current humanoids such as the Unitree G1 operate on learned policies that combine demonstration data with reinforcement learning (Unitree Robotics, 2024). Adapting these foundational-model-based controllers, such as NVIDIA’s GR00T (NVIDIA Corporation, 2025a), to reduced gravity remains an open problem and one of the key current challenges in the field (Gu et al., 2025). Recent advances in high-fidelity physics engines, such as NVIDIA Isaac Sim 5.0 with GPU-accelerated parallel computation (NVIDIA Corporation, 2024; NVIDIA Corporation, 2025d), now allow more realistic training of humanoid policies. This paper addresses the gap by applying reinforcement learning to adapt the Unitree G1’s walking policy for lunar gravity, examining both gait adaptation and the efficiency of transfer learning in a modern simulation environment.

3 Training a Lunar Unitree G1 Humanoid Robot Policy

3.1 Setting Up Isaac Sim 5.0

Installing Isaac Sim Lab in a Conda Environment

Isaac Lab was installed inside of a Conda environment using pip, as a requirement to use Unitree RL Lab Isaac Lab. Once the installation was verified Unitree RL Lab was installed. After installation, the necessary Unitree USD model files were cloned from HuggingFace and linked to the model directory. The installation was

confirmed by listing the available environments and running a sample training task, which demonstrated that the integration between Unitree RL Lab and Isaac Lab was functioning as intended (Unitree Robotics, 2025; NVIDIA Corporation, 2025a).

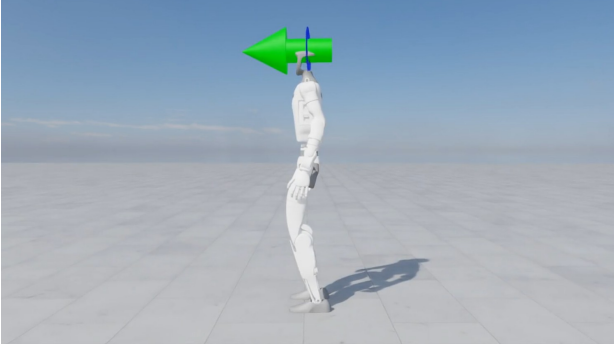


Figure 6: Custom Analysis Configuration View.

3.2 Modifying Unitree RL Lab Examples

The Unitree RL Lab features a training task called “Unitree-G1-29dof-Velocity”, which trains a G1 humanoid robot to walk on flat ground. The task provides pre-configured simulator scene, actions, rewards, observations, curriculums, commands and events that allow G1 to learn how to walk from scratch using reinforcement learning. For the Earth control policy no configurations were modified, except for the camera. The camera was configured to capture videos during the training process, configuring it to target the robots so that they are in frame every clip. Captures were typically configured to occur every 5000 global steps, except for the Earth control policy which was configured with 50000 global steps between and. To train the robot in lunar gravity the simulation gravity was then set using the simulation environment configuration. The core of the reinforcement learning configurations remain unchanged from Unitree RL Lab.

For an equal and meaningful comparison between models, a custom configuration was created to linearly increase the velocity of the robot in on direction, whilst watching the robot side on as in figure 6. This required configuring the camera to track the robot by changing the cameras origin to the robot’s inertial frame and forcing the robot to walk in one direction, requiring the “reset_root_state.uniform” event to be modified to ensure the robot always resets in the same direction. To ensure the robot to walks in a straight line with increasing velocity, which represented by the green arrow above the robot’s head in figure 6, the training curriculums were removed, and a new curriculum was created which increased the velocity command by 20% at the end of every training episode. This custom configuration allowed for

the creation of consistent videos, making the comparison between models quicker, easier and more accurate.

3.3 Training

There were three policies that were trained using reinforcement learning, the first was a control, training the G1 to walk in Earth gravity, the second was trained in Lunar gravity, continuing training from the Earth policy and the final policy was trained in straight lunar gravity. The Earth policy was used to verify the Unitree RL library, ensuring the robots do learn to walk establish a baseline for performance and to give a gauge as to how long training the rest of the policies might take. The second policy fine-tunes the Earth policy to walk in lunar gravity, aiming to investigate the performance and effectiveness of fine-tuning an Earth based policy, whilst representing a scenario where a robot may have well Earth policy and is looking to use it on the moon. The final policy was trained in lunar gravity right from the beginning, investigating the time taken to train a lunar gravity policy to compare to the time taken to fine-tune an existing model. Additionally to investigating the effects of training a policy for walking in lunar gravity has on walking gait and speed, training the three policies in this way allowed insight into whether it’s best to fine-tune a policy or start from scratch for any gravity levels the robot may want to be operated in.

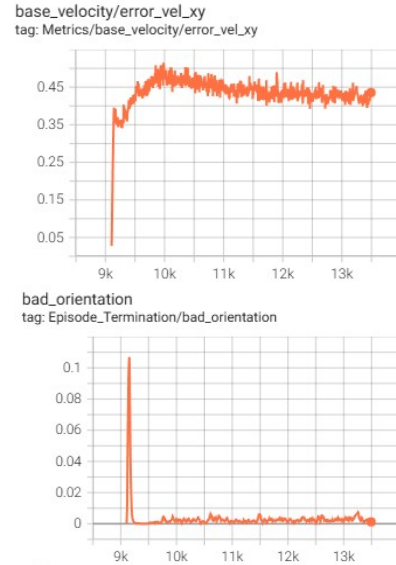


Figure 7: Velocity Error and Bad Orientation Terminations Converging on fine-tune policy.

All training was completed using the same Unitree RL Lab task, only changing the gravity between sessions and ending the training only once the average velocity error

and the average of the terminations due to bad orientation have converged. Figure 7 shows these metrics across the entire fine-tuning run, showing how they were converging, signifying the end of the training.

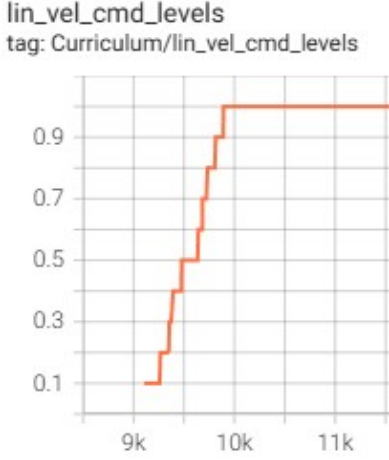


Figure 8: Velocity command increasing on fine-tune policy

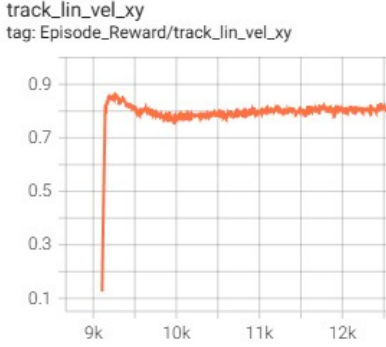


Figure 9: Reward Term for Tracking the Velocity Command.

Each policy was trained using 4096 robots, with an episode length of 20 seconds, with a series of curriculums to increase the velocity as the robots learn to walk better. A curriculum takes a range of allowable (x,y) velocities and commands a velocity within that range that is based off how far the robot walked in the last episode, meaning the shorter the distance that the robot walks, the slower the velocity requested by the curriculum. The range is bound by set limits and can be negative or positive. The range of the velocity requested was increased as shown in Figure 8 when the robots on average were able to track their requested velocity greater than 80% as demonstrated by the converging line at 0.8 in Figure 9 of the time. This combination of curriculums allows the robot to learn a policy which can handle all velocities up to the max and in any direction.

4 Results

4.1 Gravity’s Effect on Walking Gait

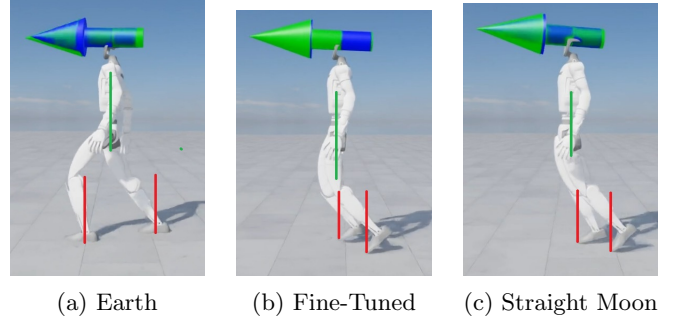
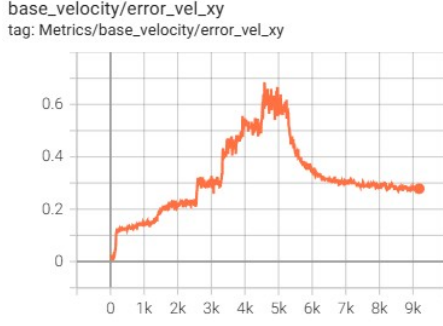


Figure 10: Comparison of walking gaits for policies trained under Earth and lunar gravity. The lunar-trained models show shorter stride lengths, a forward-shifted centre of mass, and increased reliance on toe contact compared to the Earth baseline, consistent with prior analyses of reduced-gravity locomotion.

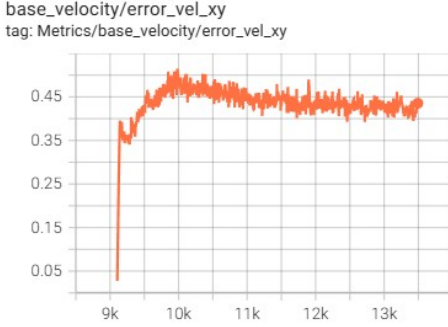
Gravity significantly altered the walking gait of the G1. When comparing the images in figure 10, the red lines are positioned at the same point on each foot illustrating the step length, and the green line represents where the centre of mass of the robot is. When looking at the Earth trained policy, the wide red lines show a long stride, with the centre of mass aligning just slightly forward of the mid-point between the red lines. Both Moon policies have a much shorter stride when compared to the Earth policy and the centre of mass has moved far forward, now roughly lining up front foots toes. This align with past research, showing that as gravity is decrease, to achieve a fast stable walk the vertical centripetal force produced by the legs must be managed, hence the weird tip toeing and minimal thigh movement, all in the effort to minimise the centripetal caused by leg movement, resulting in the shorter walking gait on their toes. The centre of mass being more forward on the Moon compared to Earth, indicates that since the acceleration of is less then Earth; to achieve the same forward acceleration vector as earth, the centre of mass must lean more forward of the bodies centre of rotation. Using modern simulation with detailed robot models, confirmed previous research such as (Omer et al., 2011, 2014) and allowed demonstrated is possible with a humanoid robot with high degrees-of-freedom.

4.2 Velocity Error

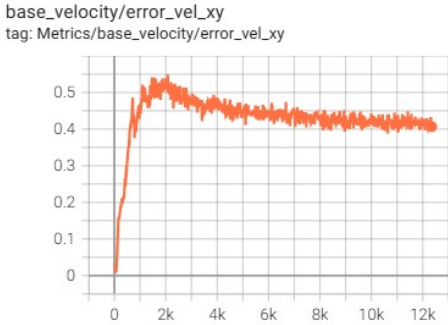
The velocity error is the mean of the error between the desired velocity and the robots actual velocity. This measurement can be used to assess the models control accuracy. When comparing these in figure 11, both



(a) Earth Velocity Error



(b) Fine-Tuned Velocity Error



(c) Moon Velocity Error

Figure 11: Walking gaits produced by fine-tuned and lunar-from-scratch policies. Both exhibit similar adaptations to reduced gravity, but the fine-tuned policy achieves equivalent gait stability with significantly reduced training time, demonstrating the efficiency of transfer learning.

lunar policies have roughly 0.4 error, 0.1 above the velocity error for the earth policy. Since the policies were trained until they converged, using the same reinforcement learning task configuration, the difference in error can only be accounted for by the reduction of gravity. However, this also appears to help the robots learn walk more without falling. Looking at the graph for the Earth policy training, the velocity error steadily increases over iterations before converging compared to the lunar policies straight shot up. This may suggest the robot’s aren’t falling over as quickly allowing them to learn how to hold

their weight quicker.

4.3 Training Times

Table 2: Training Times.

Policy	Earth	Fine-Tune	Moon
Iteration	9,100	4,300	12,300
Time (Hours)	14.5	7	20

When comparing the time taken to train the 2 lunar policies with their results, it shows if try to adapt a Earth based humanoid control policy, fine-tuning it is more effective then training a new one from scratch. Looking the training times in table 2, the fine-tuning only took 7 hours to achieve a policy that is as capable as the one trained from scratch which took 20 hours. When comparing the walking gaits in figure 10, the walking gaits of the two moon policies look almost identical. Both policies perform similarly as Figure 11 shows both of their velocity errors begin to converge around 0.4. When working with an existing humanoid robot policy trained for Earth gravity, it is faster to fine-tune to the existing policy to Lunar gravity, then to again start from scratch, while offering same level of performance.

5 Discussion

The results of this study highlight both the challenges and opportunities of adapting humanoid locomotion policies to reduced gravity environments using reinforcement learning. Training revealed clear differences in gait characteristics between Earth and lunar gravity, consistent with prior simulation-based analyses of low-gravity locomotion Omer et al. (2011) and Omer et al. (2014). Specifically, lunar-trained policies exhibited shortened stride lengths, toe-dominant walking, and a more forward-shifted centre of mass. These adjustments suggest that reinforcement learning is capable of autonomously identifying compensatory strategies to maintain stability in reduced gravity, without requiring explicit human-designed gait rules.

A key finding is the efficiency of fine-tuning an Earth-trained policy compared with training from scratch. Fine-tuned models achieved equivalent performance to lunar-from-scratch models in less than half the training time, supporting the notion that prior knowledge embedded in baseline policies can accelerate adaptation across environments. This insight has direct implications for future planetary robotics missions, where computational resources may be limited and rapid policy adaptation is critical.

At the same time, the analysis of velocity errors shows that policies trained in lunar gravity do not yet achieve

the same control precision as their Earth-trained counterparts. Although convergence was achieved, residual error levels remained higher under lunar conditions, suggesting that reward function design and curriculum shaping will be important areas for refinement. Adjustments such as weighting penalties for gait instability or experimenting with alternative reward structures may yield policies with reduced error and more efficient locomotion.

These findings also raise broader considerations for humanoid deployment in extraterrestrial habitats. While walking on flat simulated terrain demonstrates feasibility, real lunar surfaces are characterised by uneven regolith, dust, and reduced friction conditions. Extending training to incorporate such environmental complexities will be essential to ensure robustness. Furthermore, combining reinforcement learning with foundational humanoid robot models could provide generalisable locomotion strategies that transfer more seamlessly across gravity levels and terrains.

Overall, this study demonstrates that reinforcement learning in high-fidelity physics simulators can not only replicate earlier analytical insights into low-gravity locomotion but also generate practical, robot-specific walking policies. By validating that fine-tuning is both faster and equally effective as training from scratch, the work provides a foundation for scalable approaches to humanoid adaptation in planetary exploration scenarios.

Despite encouraging results, this study has several limitations. First, all experiments were conducted in simulation using a physics engine; the sim-to-real gap remains a major challenge, and transferring lunar-trained policies to physical hardware will likely expose unmodelled dynamics such as actuator non-linearities, delays, and compliance. Second, the training environment assumed flat terrain and stable contact, whereas lunar regolith is uneven, dusty, and characterised by reduced friction. Third, only forward walking was considered; other gaits such as turning, climbing, or recovery from perturbations were not addressed. Finally, computational cost was non-trivial, with training runs requiring multiple days of GPU compute, which may limit scalability in resource-constrained mission settings. Addressing these limitations will be key to advancing from proof-of-concept to deployable systems.

6 Conclusion

This paper has presented a reinforcement learning approach for training humanoid robots to walk in simulated lunar gravity, using the Unitree G1 as a case study. Results demonstrate that policies trained in low gravity autonomously adapt gait strategies, including shorter stride lengths and altered centre of mass positioning, echoing prior analytical studies while revealing robot-

specific compensations. Fine-tuning an Earth-trained model proved highly effective, achieving performance comparable to lunar-from-scratch training in less than half the time, underscoring the value of transfer learning across environments.

These findings contribute to the growing body of work on humanoid robotics for space exploration, showing that scalable and adaptable locomotion policies can be developed for extraterrestrial environments.

In summary, the key contributions of this work are:

- Demonstrating that reinforcement learning can autonomously adapt humanoid locomotion policies to lunar gravity, producing gait strategies consistent with prior theoretical predictions.
- Showing that transfer learning via fine-tuning reduces training time substantially while maintaining performance, offering a practical pathway for planetary robotics missions.
- Extending reinforcement learning studies on humanoid locomotion beyond Earth-based benchmarks to extraterrestrial contexts, establishing a foundation for scalable deployment in space exploration.

6.1 Future Work

Further investigation into altering of the reinforcement learning rewards, increasing the training time drastically and understanding how to utilise a policy to generate synthetic robot demonstration data. Though manipulating the reward weights, it could be possible to discover more novel walking gaits, possibly yielding in a new style of locomotion. These discoveries could be aided by increasing the training time until all metrics have coverage. Generating syntactic data using a novel walking gait policy could allow foundation model like NVIDIA’s Isaac Groot (NVIDIA Corporation, 2025b), the ability to move around more effectively in any gravity. Further work in these areas could unlock insights that would allow Unitree G1’s to walk on the moon in lunar habitat, in its own strange but efficient way.

References

- Cavagna, G. A., Willems, P. A., and Heglund, N. C. (2000). The role of gravity in human walking: pendular energy exchange, external work and optimal speed. *The Journal of Physiology*, 528(3):657–668.
- Chen, P., Wang, Y., Luo, C., Cai, W., and Zhao, M. (2020). Hifar: Multi-stage curriculum learning for high-dynamics humanoid fall recovery. *Engineering Applications of Artificial Intelligence*. Forthcoming or preprint.
- DeepMind (2021). Mujoco is now free and open source. <https://deepmind.com/blog/article/mujoco>. Accessed: 2025-09-21.

- Gu, Z., Li, J., Shen, W., Yu, W., Xie, Z., McCrory, S., Cheng, X., Shamsah, A., Griffin, R., Liu, C. K., Kheddar, A., Peng, X. B., Zhu, Y., Shi, G., Nguyen, Q., Cheng, G., Gao, H., and Zhao, Y. (2025). Humanoid locomotion and manipulation: Current progress and challenges in control, planning, and learning.
- Isaac Sim Development Team (2025). Isaac sim on github. <https://github.com/isaac-sim/IsaacSim>. Accessed 2025.
- Kee, D., Wyeth, G., and Roberts, J. (2004). Biologically inspired joint control for a humanoid robot. In *4th IEEE/RAS International Conference on Humanoid Robots, 2004.*, volume 1, pages 385–401 Vol. 1.
- Kober, J., Bagnell, J. A., and Peters, J. (2013). Reinforcement learning in robotics: A survey. *The International Journal of Robotics Research*, 32(11):1238–1274.
- Liu, S., Wu, L., Tan, H., Chen, H., Zhu, J., et al. (2024). Rdt-1b: a diffusion foundation model for bimanual manipulation. *arXiv preprint arXiv:2410.07864*.
- NVIDIA Corporation (2024). Nvidia isaac sim. Accessed: 2025-04-09. Specific version updates may alter the year.
- NVIDIA Corporation (2025a). Isaac gr00t n1: An open foundation model for humanoid robots. *arXiv preprint arXiv:2503.14734*.
- NVIDIA Corporation (2025b). Isaac gr00t n1.5 github repository.
- NVIDIA Corporation (2025a). Isaac lab: Unified framework for robot learning. Technical documentation. Isaac Sim Documentation – Includes APIs, environments, GPU-accelerated simulation, domain randomisation, and robot models from companies such as Unitree, Boston Dynamics, and Franka Emika.
- NVIDIA Corporation (2025b). Isaac sim requirements — isaac sim documentation. <https://docs.isaacsim.omniverse.nvidia.com/5.0.0/installation/requirements.html>. Accessed 2025.
- NVIDIA Corporation (2025c). Known issues — isaac sim documentation. https://docs.robotsfan.com/isaacsim/5.0.0/overview/known_issues.html. Accessed 2025.
- NVIDIA Corporation (2025d). Physics — isaac sim documentation. <https://docs.isaacsim.omniverse.nvidia.com/latest/physics/index.html>. Accessed 2025.
- Omer, A., Hashimoto, K., ok Lim, H., and Takanishi, A. (2011). Initial study of bipedal robot locomotion approach on different gravity levels. In *International Symposium on Artificial Intelligence, Robotics and Automation in Space (i-SAIRAS)*.
- Omer, A., Hashimoto, K., ok Lim, H., and Takanishi, A. (2014). Study of bipedal robot walking motion in low gravity: Investigation and analysis. *International Journal of Advanced Robotic Systems*, 11:139.
- Radford, N. A., Strawser, P., Hambuchen, K., Mehling, J. S., Verdeyen, W. K., Donnan, A. S., Holley, J., Sanchez, J., Nguyen, V., Bridgwater, L., Berka, R., Ambrose, R., Diftler, M., Bluethmann, W., Paine, N., Kuehn, J., Beeson, P., Ballard, B., Davis, D., Tyree, K., Kim, S.-H., Lim, B., Gaddis, B., Ames, A., Faber, J., Strawser, P., Aldridge, H., Askew, R. S., Burrige, R. R., Bluethmann, W., Lovchik, C., Magruder, D., and Rehnmark, F. (2015). Valkyrie: Nasa’s first bipedal humanoid robot. *Journal of Field Robotics*, 32:397–419.
- Seo, Y., Li, S., Lee, K., Abbeel, P., and Ba, J. (2025). Fasttd3: Simple, fast, and capable reinforcement learning for humanoid control. *arXiv preprint arXiv:2505.22642*.
- Sferrazza, C., Wang, Y., Krishnan, S., Shi, Y., and Fazeli, N. (2024). Humanoidbench: Simulated humanoid benchmark for whole-body locomotion. *arXiv preprint arXiv:2403.10506*.
- Singh, R. P., Benallegue, M., Morisawa, M., Cisneros, R., and Kanehiro, F. (2022). Learning bipedal walking on planned footsteps for humanoid robots. *arXiv preprint arXiv:2207.12644*.
- Stoica, A., Keymeulen, D., Csaszar, A., Gan, Q., Hidalgo, T., Moore, J., Newton, J., Sandoval, S., and Xu, J. (2005). Humanoids for lunar and planetary surface operations. In *5th IEEE-RAS International Conference on Humanoid Robots*, pages 1–6.
- Tassa, Y., Doron, Y., Muldal, A., Erez, T., Li, Y., Casarini, F., Kumar, S., King, M., Kohli, P., Lefrancq, A., Lillicrap, T., Galashov, A., and Silver, D. (2018). Deepmind control suite. *arXiv preprint arXiv:1801.00690*.
- Tesla (2022). Tesla ai day 2022. YouTube.
- Todorov, E., Erez, T., and Tassa, Y. (2012). Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033. IEEE.
- Unitree Robotics (2024). Unitree rl lab. Accessed: 2025-09-19.
- Unitree Robotics (2025). Unitree rl lab: Reinforcement learning implementation for unitree robots, based on isaacsim. Accessed 20 September 2025.